

Strategy Brief #4

Coastal Infrastructure Exposure



Assessing how the growing concentration of critical infrastructure (seaports, airports, energy, water systems) in low-lying coastal zones is creating compounding economic and operational risks that current planning frameworks are not equipped to address.

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Introduction

Critical infrastructure (Seaport, airport, transport, water, energy) is increasingly concentrated in **coastal urban areas**.



Key Challenge

Infrastructure systems are highly exposed to **flooding, subsidence, and environmental degradation**, with limited resilience planning.

Strategic Insights

- Infrastructure development often **does not account for long-term risks.**
- Exposure is increasing due to **urban expansion and asset concentration.**
- Failure of infrastructure can trigger **system-wide disruptions.**

Strategic Implications

- High risk of **economic losses and service disruptions.**
- Increased maintenance and replacement costs.
- Reduced attractiveness for **investment and development.**

Recommendations

- Integrate **risk assessments** into **infrastructure planning**.
- Prioritise **resilient and adaptive design standards**.
- Develop **asset-level risk monitoring systems**.
- Align infrastructure investment with **climate adaptation strategies**.



Value Proposition

Resilient infrastructure reduces **long-term costs** and **enhances economic stability**.